



**SIMATS**  
ENGINEERING



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

# **SMART RESUME ANALYSIS AND JOB MATCHING SYSTEM USING NATURAL LANGUAGE PROCESSING**

## **A CAPSTONE PROJECT REPORT**

**A Capstone Project Report submitted in partial fulfilment of the requirements for  
the Course of  
DSA0306 – NATURAL LANGUAGE PROCESSING  
to the award of the degree of  
BACHELOR OF TECHNOLOGY  
IN  
ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING**

**Submitted by**

**P. MUNI KARTHIK – 192425061**

**Y. SRAVAN KUMAR – 192472289**

**K. KALYAN – 192425301**

**N. VENKATA SIVA – 192472236**

**Under the Supervision of**

**KUMARAGURUBARAN T**

**SIMATS ENGINEERING**

**Saveetha Institute of Medical and Technical Sciences  
Chennai-602105**

**SEPTEMBER - 2026**

## **SIMATS ENGINEERING**

**Saveetha Institute of Medical and Technical Sciences**

**Chennai-602105**

### **BONAFIDE CERTIFICATE**

This is to certify that the Capstone Project entitled “Smart Resume Analysis and Job Matching System Using Natural Language Processing” has been carried out by P. MUNI KARTHIK (192425061), Y. SRAVAN KUMAR (192472289), K. KALYAN (192425301), N. VENKATA SIVA (192472236) under the supervision of KUMARAGURUBARAN T and is submitted in partial fulfilment of the requirements for the current semester of the B.Tech – AIML Engineering program at Saveetha Institute of Medical and Technical Sciences, Chennai.

#### **COURSE COORDINATOR**

Professor

Department of CSE,

Saveetha School of Engineering

SIMATS, Chennai

#### **COURSE FACULTY**

Dr. T. KUMARAGURABAN

Professor

Department of CSE,

Saveetha School of Engineering,

SIMATS, Chennai

# **SIMATS ENGINEERING**

**Saveetha Institute of Medical and Technical Sciences**

**Chennai-602105**

## **DECLARATION**

We, P. MUNI KARTHIK, Y. SRAVAN KUMAR, K. KALYAN and N. VENKATA SIVA of the AIML, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Smart Resume Analysis and Job Matching System Using Natural Language Processing” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged.

We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

Date: \_\_\_\_\_

Signature of Students:

P. Muni Karthik – 192425061

Y. Sravan Kumar – 192472289

K. Kalyan – 192425301

N. Venkata Siva – 192472236

## SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai-602105

### INDIVIDUAL CONTRIBUTION STATEMENT

| Student                      | Specific Responsibilities                                       | Design & Development                                                            | Testing & Analysis                                               | Report Contribution                                        | Approx. (%) |
|------------------------------|-----------------------------------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------------------------|-------------|
| P. Muni Karthik<br>192425061 | Part 1 – Resume Ingestion & NLP Preprocessing                   | Resume text extraction; tokenization; stop-word removal; clean-text preparation | Validation of extracted and preprocessed resume text             | Chapters 1–5, preprocessing, testing and reflection        | 25%         |
| Y. Sravan Kumar<br>192472289 | Part 2 – NLP Skill Extraction                                   | TF-IDF implementation; term-frequency analysis; skill-feature identification    | Verification of important terms and skill weights                | Chapters 1–5, skill extraction, analysis and reflection    | 25%         |
| K. Kalyan<br>192425301       | Part 3 – Semantic Resume–Job Matching                           | SBERT embeddings; contextual representation; cosine similarity integration      | Verification of semantic similarity and match scores             | Chapters 1–5, semantic matching, testing and reflection    | 25%         |
| N. Venkata Siva<br>192472236 | Part 4 – Job Recommendation, Performance Analysis & Integration | Job ranking; recommendation interface; system integration and analytics         | Testing recommendations, execution performance and final outputs | Chapters 1–5, performance analysis, testing and reflection | 25%         |
| TOTAL                        |                                                                 |                                                                                 |                                                                  |                                                            | 100%        |

# ABSTRACT

Finding suitable jobs based on candidate skills is challenging because resumes and job descriptions contain large amounts of unstructured textual information. Traditional recruitment systems often rely mainly on exact keyword matching, which may miss semantically related skills and can produce less relevant recommendations.

The proposed Smart Resume Analysis and Job Matching System uses Natural Language Processing (NLP) techniques to analyze resumes and job descriptions and generate relevant job recommendations. Resume text is extracted and preprocessed using tokenization and stop-word removal. TF-IDF is then used to identify important terms and skill-related features. Sentence-BERT (SBERT) converts candidate and job-description text into contextual semantic embeddings, while cosine similarity measures the similarity between the resulting vectors.

The system combines these techniques as a sequential pipeline rather than comparing algorithms against each other. The resulting compatibility scores are used to rank available jobs and present suitable recommendations. The implementation also provides analytical views of extracted skills, semantic embeddings, compatibility scores and system performance.

The project demonstrates how multiple NLP techniques can be integrated into a practical resume screening and job matching application. The individual contributions collectively cover text processing, feature extraction, semantic matching, recommendation, testing and system integration.

**Keywords:** Natural Language Processing; Resume Analysis; TF-IDF; Sentence-BERT; SBERT; Cosine Similarity; Semantic Matching; Job Recommendation; Text Preprocessing; Embeddings

## TABLE OF CONTENTS

| Sl. No | Title                                                                                       | Page No |
|--------|---------------------------------------------------------------------------------------------|---------|
| i      | CERTIFICATE FROM THE SUPERVISOR                                                             | ii      |
| ii     | DECLARATION BY THE CANDIDATES                                                               | iii     |
| iii    | INDIVIDUAL CONTRIBUTION STATEMENT                                                           | iv      |
| iv     | ABSTRACT                                                                                    | v       |
| v      | LIST OF FIGURES                                                                             | vi      |
| vi     | LIST OF TABLES                                                                              | vii     |
| vii    | LIST OF ABBREVIATIONS / SYMBOLS                                                             | viii    |
| 1      | CHAPTER 1: Introduction and Engineering Problem                                             | 1       |
| 2      | CHAPTER 2: Literature Review and Existing Solutions                                         | 4       |
| 3      | CHAPTER 3: Engineering Design & Trade-offs, Sustainability, Ethics, Safety, Risk & Security | 7       |
| 4      | CHAPTER 4: Implementation, Testing & Results, Project Management                            | 12      |
| 5      | CHAPTER 5: Conclusion, Future Work and Reflection                                           | 18      |
|        | References                                                                                  | 19      |
|        | Appendices                                                                                  | 20      |

## LIST OF TABLES

| Table No. | Table Name                                                 |
|-----------|------------------------------------------------------------|
| Table 1   | Individual Contribution Statement                          |
| Table 2   | Project Objectives                                         |
| Table 3   | Stakeholder and User Requirements                          |
| Table 4   | Functional and Non-Functional Requirements                 |
| Table 5   | Design Specifications                                      |
| Table 6   | Alternative Solution Evaluation                            |
| Table 7   | Trade-off Analysis                                         |
| Table 8   | Sustainability, Ethics, Safety and Security Considerations |
| Table 9   | Risk Register                                              |
| Table 10  | Software Tools and Resources                               |
| Table 11  | Test Plan and Verification                                 |
| Table 12  | Objective Achievement                                      |
| Table 13  | Project Milestones                                         |
| Table 14  | Team Responsibilities                                      |
| Table 15  | Project Budget                                             |

## LIST OF FIGURES

| Figure No. | Figure Name                                           |
|------------|-------------------------------------------------------|
| Figure 1   | Smart resume analysis and job matching system flow    |
| Figure 2   | Resume ingestion and extracted entities interface     |
| Figure 3   | TF-IDF skills and feature extraction interface        |
| Figure 4   | SBERT semantic job matching engine                    |
| Figure 5   | Candidate compatibility and job recommendation result |

## LIST OF ABBREVIATIONS / SYMBOLS

| Symbol / Term     | Description                                             |
|-------------------|---------------------------------------------------------|
| NLP               | Natural Language Processing                             |
| TF-IDF            | Term Frequency–Inverse Document Frequency               |
| SBERT             | Sentence-BERT                                           |
| BERT              | Bidirectional Encoder Representations from Transformers |
| UI                | User Interface                                          |
| UX                | User Experience                                         |
| I/O               | Input / Output                                          |
| PDF               | Portable Document Format                                |
| DOCX              | Microsoft Word Open XML Document                        |
| TXT               | Text File                                               |
| API               | Application Programming Interface                       |
| ML                | Machine Learning                                        |
| AI                | Artificial Intelligence                                 |
| Embedding         | Numerical vector representation of text                 |
| Cosine Similarity | Measure of similarity between two vectors               |
| Vector            | Numerical representation of a text profile              |
| Dim               | Embedding vector dimension                              |



# CHAPTER 1

## INTRODUCTION AND ENGINEERING PROBLEM

### 1.1 Background

Online recruitment platforms contain a large number of job postings, while candidates submit resumes containing information about skills, education and experience. Resumes are commonly stored as unstructured documents, making automated analysis difficult when the information is distributed across different sections and expressed using different terms.

Natural Language Processing provides methods for processing and analyzing textual information. In resume screening, NLP can be used to extract relevant information, identify candidate skills, analyze job descriptions and determine the semantic relationship between a candidate profile and a job requirement.

The proposed system integrates multiple NLP techniques in a functional pipeline. Tokenization and stop-word removal prepare the text, TF-IDF identifies important terms, SBERT generates contextual semantic representations, and cosine similarity calculates the degree of similarity between candidate and job vectors. The combined result is used to rank and recommend suitable jobs.

### 1.2 Need for the Project

The project is needed to provide an automated method for analyzing resumes and matching candidates with relevant job opportunities. Manual comparison of resumes against many job descriptions is time-consuming and difficult to scale.

Exact keyword matching can also fail when related skills are expressed using different terminology. A semantic NLP-based approach can capture contextual relationships and provide more meaningful matching between candidate profiles and job descriptions.

### 1.3 Problem Statement

The engineering problem is to develop and validate an NLP-based system that receives resume documents and job descriptions, extracts meaningful candidate information, identifies important skills, represents candidate and job profiles semantically, calculates compatibility and ranks suitable jobs.

The system must process unstructured resume text, preserve relevant technical terms, identify job requirements and generate a reliable matching score.

## 1.4 Aim

To develop and validate a smart resume analysis and job matching system using Natural Language Processing that extracts candidate information, identifies relevant skills, captures semantic relationships between resumes and job descriptions, calculates compatibility scores and recommends suitable job opportunities.

## 1.5 Objectives

| Objective               | Measurable Outcome                                                                         |
|-------------------------|--------------------------------------------------------------------------------------------|
| Resume Text Extraction  | Extract text from PDF/DOCX/TXT resume documents.                                           |
| NLP Preprocessing       | Tokenize text and remove common stop words while preserving relevant technical terms.      |
| Skill Extraction        | Use TF-IDF to identify important terms and skill-related resume features.                  |
| Semantic Representation | Generate contextual resume and job embeddings using SBERT.                                 |
| Similarity Calculation  | Calculate semantic similarity using cosine similarity.                                     |
| Job Recommendation      | Rank jobs using compatibility scores and recommend suitable opportunities.                 |
| System Verification     | Test extraction, preprocessing, skill identification, matching and recommendation outputs. |

## 1.6 Scope and Expected Outcome

The system covers resume document ingestion, text extraction, NLP preprocessing, skill-oriented feature extraction, job-description analysis, SBERT-based semantic representation, cosine-similarity scoring, job ranking and recommendation. The system is intended as an academic prototype and does not claim to replace professional recruitment decisions.

## CHAPTER 2

### LITERATURE REVIEW AND EXISTING SOLUTIONS

#### 2.1 Review Method

The review considered common NLP and information-retrieval approaches used in resume screening and job matching. The comparison focused on text representation, keyword importance, contextual understanding, similarity calculation and suitability for the proposed recruitment problem. Particular attention was given to keyword matching, TF-IDF, transformer-based embeddings and cosine similarity.

#### 2.2 Review of Existing Work

Traditional recruitment systems frequently use keyword-based search to retrieve jobs containing terms similar to those in a candidate resume. Although this approach is simple, it may fail when a candidate and a job description use related but non-identical expressions. TF-IDF provides a statistical representation of term importance by considering term frequency and inverse document frequency.

| Existing Approach     | Advantages                                                | Limitations                                       | Relevance to the System               |
|-----------------------|-----------------------------------------------------------|---------------------------------------------------|---------------------------------------|
| Keyword Matching      | Simple and fast; easy to implement                        | Exact terms may miss semantically related skills  | Baseline concept                      |
| TF-IDF                | Highlights important terms statistically                  | Limited contextual understanding                  | Selected for skill/feature extraction |
| Word-level Embeddings | Represent words numerically                               | Context and sentence-level meaning may be limited | Possible alternative                  |
| SBERT                 | Captures contextual semantic meaning; sentence embeddings | Higher computational requirement                  | Selected for semantic matching        |
| Proposed NLP Pipeline | Combines complementary NLP stages                         | Requires integration and validation               | Selected for academic system          |

## 2.3 Engineering Gap

The engineering gap addressed in this project is the need for an integrated and explainable workflow that can process unstructured resumes, identify important skills and compare candidate profiles with job descriptions semantically. A system based only on exact keywords may miss relevant relationships, while a semantic model alone does not explicitly provide the term-importance information required for skill-oriented analysis.

The proposed system addresses this gap by assigning each NLP technique a specific role in the overall pipeline. This avoids unnecessary algorithm comparison and instead focuses on how complementary methods contribute to the final job-matching objective.

## 2.4 Proposed Contribution

The proposed contribution is an integrated Smart Resume Analysis and Job Matching System designed to automate the analysis of candidate resumes and identify suitable job opportunities using Natural Language Processing techniques. The system is developed as a sequential and modular pipeline in which each NLP technique performs a specific function within the overall job-matching process. Rather than treating the algorithms as competing alternatives, the proposed approach combines their individual strengths to achieve effective resume analysis and semantic job recommendation.

The workflow begins with resume document ingestion and text extraction, where resumes in supported formats such as PDF, DOCX and TXT are processed to obtain the raw textual content. The extracted resume text is then subjected to NLP preprocessing, including tokenization and stop-word removal, to reduce unnecessary textual information and prepare the content for subsequent analysis. This stage ensures that the important technical and professional terms present in the candidate's resume are retained for further processing.

Following preprocessing, the system applies TF-IDF (Term Frequency–Inverse Document Frequency) to identify and assign importance to significant terms appearing in the resume. TF-IDF provides a numerical weight for words based on their frequency within the resume and their occurrence across the available document corpus. In the proposed system, this stage is particularly useful for identifying high-signal technical skills and other important terms that contribute to the candidate profile. The extracted features can therefore be used to understand the lexical strengths of the candidate and support the matching process.

## CHAPTER 3

### ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

#### 3.1 Requirements

| Stakeholder             | Requirement                                                               |
|-------------------------|---------------------------------------------------------------------------|
| Candidate               | Upload a resume and receive relevant job recommendations.                 |
| Recruiter / System User | Obtain structured candidate information and compatibility results.        |
| Project Evaluator       | Verify correct implementation of the NLP pipeline.                        |
| System Developer        | Process resume and job-description text using appropriate NLP techniques. |

| Requirement Type       | Measurable Target                                          |
|------------------------|------------------------------------------------------------|
| Resume processing      | Resume text is extracted from supported document formats.  |
| Preprocessing          | Text is tokenized and common stop words are removed.       |
| Skill extraction       | Important skill-related terms are identified using TF-IDF. |
| Semantic matching      | Resume and job descriptions are represented using SBERT.   |
| Similarity calculation | A cosine similarity score is generated.                    |
| Recommendation         | Jobs are ranked according to matching scores.              |
| Correctness            | Outputs are verified using predefined test cases.          |

### 3.2 Constraints & Applicable Practices

The proposed system is constrained by the quality and structure of resume and job-description text, the available skill vocabulary and the computational requirements of transformer-based embeddings. The system should preserve important technical terms during preprocessing and should avoid removing words that contribute to skill identification.

TF-IDF provides statistical term importance, while SBERT provides contextual semantic representations. The techniques are therefore used as complementary components. Correctness is maintained through explicit preprocessing steps, controlled feature extraction, similarity calculation and verification using representative test cases.

### 3.3 Design Specifications

The design specification is based on a sequential NLP processing pipeline. Resume documents and job descriptions are supplied as inputs. The system extracts raw text, preprocesses it, identifies important skill-related terms, generates contextual embeddings and calculates similarity before ranking job opportunities.

| Parameter           | Target / Requirement              | Unit      | Priority |
|---------------------|-----------------------------------|-----------|----------|
| Input formats       | PDF / DOCX / TXT                  | Document  | High     |
| Preprocessing       | Tokenization + Stop-word removal  | Text      | High     |
| Skill extraction    | TF-IDF feature weighting          | Score     | High     |
| Semantic model      | Sentence-BERT                     | Embedding | High     |
| Embedding dimension | 768-dimensional contextual vector | Dimension | High     |
| Similarity          | Cosine similarity                 | Score     | High     |
| Output              | Ranked job recommendations        | Jobs      | High     |

### 3.4 Alternative Solutions & Evaluation

| Criterion           | Keyword Matching | TF-IDF | SBERT  | Proposed Pipeline |
|---------------------|------------------|--------|--------|-------------------|
| Exact term matching | High             | High   | Low    | Balanced          |
| Term importance     | Low              | High   | Low    | High              |
| Contextual meaning  | Low              | Medium | High   | High              |
| Semantic matching   | Low              | Medium | High   | High              |
| Explainability      | High             | High   | Medium | High              |
| System suitability  | Medium           | High   | High   | High              |

### 3.5 Selected Design & Detailed Design

The selected design uses a modular NLP pipeline in which each technique performs a distinct operation. Resume text is extracted first. Tokenization and stop-word removal create cleaned text. TF-IDF then provides term-importance information for skill-oriented analysis. SBERT encodes candidate and job text into contextual vectors. Cosine similarity compares these vectors, and the resulting scores are used for job ranking.

### 3.6 Trade-off Analysis

| Design Decision                   | Alternative Considered | Benefit                                 | Disadvantage                       | Final Decision & Justification             |
|-----------------------------------|------------------------|-----------------------------------------|------------------------------------|--------------------------------------------|
| TF-IDF for skill features         | Keyword frequency only | Provides document-level term importance | Limited semantic context           | Selected for interpretable skill weighting |
| SBERT for semantic representation | TF-IDF vectors         | Captures contextual meaning             | Higher computation                 | Selected for semantic matching             |
| Cosine similarity                 | Exact overlap          | Works directly with embedding vectors   | Produces a similarity value rather | Selected for vector comparison             |

### 3.7 Sustainability, Ethics, Safety, Risk & Security

The system is designed as a lightweight academic prototype with modular processing. Ethical considerations include avoiding misleading claims that an automatically generated match score is equivalent to a hiring decision. The system should be used as decision support rather than as the sole basis for recruitment.

Safety considerations include preventing incorrect recommendations caused by poor text extraction, incomplete resumes or unsuitable matching thresholds. Security considerations include protecting uploaded resumes and avoiding unauthorized access to candidate information.

| Domain         | Consideration                                  | Evidence Evaluated              | Impact on Final Decision                    |
|----------------|------------------------------------------------|---------------------------------|---------------------------------------------|
| Sustainability | Efficient processing and reusable NLP pipeline | Modular implementation          | Reduced unnecessary computation             |
| Ethics         | Avoid automated hiring claims                  | Defined project scope           | Results interpreted as decision support     |
| Safety         | Avoid misleading recommendations               | Testing and validation          | Verification required before interpretation |
| Security       | Protect resume and candidate data              | Controlled academic environment | Suitable for prototype                      |

### Risk Register

| Risk                                         | Likelihood | Severity | Risk Level | Mitigation                                      | Residual Risk |
|----------------------------------------------|------------|----------|------------|-------------------------------------------------|---------------|
| Incorrect text extraction                    | Medium     | High     | High       | Validate extracted content                      | Low           |
| Important skill removed during preprocessing | Medium     | High     | High       | Preserve technical terms and test preprocessing | Low           |
| Incorrect TF-IDF weighting                   | Medium     | Medium   | Medium     | Verify feature weights                          | Low           |



## CHAPTER 4

### IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

#### 4.1 Development Approach & Resources

The smart resume analysis and job matching system was developed using a structured NLP-development approach. The implementation was divided into individual stages so that each processing operation could be developed and verified independently. The primary focus was on building an integrated pipeline consisting of resume ingestion, preprocessing, TF-IDF feature extraction, SBERT semantic representation, cosine similarity and job recommendation.

The development process began by identifying the inputs and outputs required by each module. The input consists of a resume document and a collection of job descriptions. The output consists of extracted candidate information, skill-related features, semantic similarity scores, compatibility results and ranked job recommendations.

| Resource                    | Purpose                                      |
|-----------------------------|----------------------------------------------|
| Python                      | Core NLP and application implementation      |
| PDF/DOCX/TXT parsers        | Resume text extraction                       |
| NLP preprocessing libraries | Tokenization and stop-word removal           |
| TF-IDF                      | Term and feature weighting                   |
| Sentence-BERT               | Contextual semantic embeddings               |
| Cosine Similarity           | Vector similarity calculation                |
| Job dataset                 | Testing job descriptions and recommendations |
| Microsoft Word              | Technical report preparation                 |

#### 4.2 Software Implementation & Modern Tools Used

The system was implemented as a modular NLP application. Python was used for application logic and integration of the NLP components. Resume files are converted into raw text before preprocessing. TF-IDF is applied for term weighting, while a Sentence-BERT model generates contextual embeddings for resume and job-description text. Cosine similarity is then used to produce compatibility scores.

| <b>Tool / Technology</b> | <b>Purpose</b>                          |
|--------------------------|-----------------------------------------|
| Python                   | Core application and NLP implementation |
| NLP libraries            | Text preprocessing and tokenization     |
| TF-IDF vectorizer        | Skill-oriented feature extraction       |
| Sentence-BERT            | Semantic embedding generation           |
| Cosine similarity        | Similarity scoring                      |
| PDF/DOCX/TXT processing  | Resume ingestion                        |
| Web UI / Dashboard       | Visualization and user interaction      |
| Microsoft Word           | Documentation and report preparation    |

### 4.3 Test Plan & Verification

| <b>Requirement</b>      | <b>Test Method</b>                          | <b>Acceptance Criterion</b>                              |
|-------------------------|---------------------------------------------|----------------------------------------------------------|
| Resume extraction       | Upload PDF/DOCX/TXT sample                  | Resume text is extracted correctly                       |
| Tokenization            | Process extracted text                      | Tokens are generated                                     |
| Stop-word removal       | Process sample text containing common words | Common stop words are removed while key terms remain     |
| TF-IDF skill extraction | Run TF-IDF over resume corpus               | Important skill-related terms receive meaningful weights |
| SBERT embedding         | Encode resume and job description           | Contextual vectors are generated                         |
| Cosine similarity       | Compare candidate and job vectors           | A similarity score is generated                          |
| Job ranking             | Evaluate multiple job descriptions          | Jobs are ordered by compatibility                        |
| Recommendation          | Display top-ranked jobs                     | Relevant jobs are presented                              |

## 4.4 Results

The implemented system successfully extracts resume text, preprocesses the text, identifies important skills and creates a candidate profile. The interface also provides a live representation of the processing pipeline and displays the extracted entities and embedding information.

The TF-IDF feature-extraction module displays the relative importance of skill-related terms. In the demonstrated execution, Python, PyTorch, NLP and Transformers, Machine Learning and SBERT & Embeddings receive high TF-IDF weights, showing how the statistical feature-extraction stage identifies strong lexical signals.

The SBERT semantic matching module generates candidate and target-job vectors and calculates a compatibility score. In the demonstrated interface, the Senior Python & AI Engineer role receives a 95.8% compatibility result. This score represents candidate-job similarity produced by the system and is not an accuracy percentage for SBERT.

The job recommendation interface ranks multiple positions according to their calculated compatibility. The demonstrated dashboard shows high-fit roles such as AI Search & Information Retrieval Specialist, Applied NLP & Text Mining and NLP Research & ML Scientist, with compatibility values above 96%.

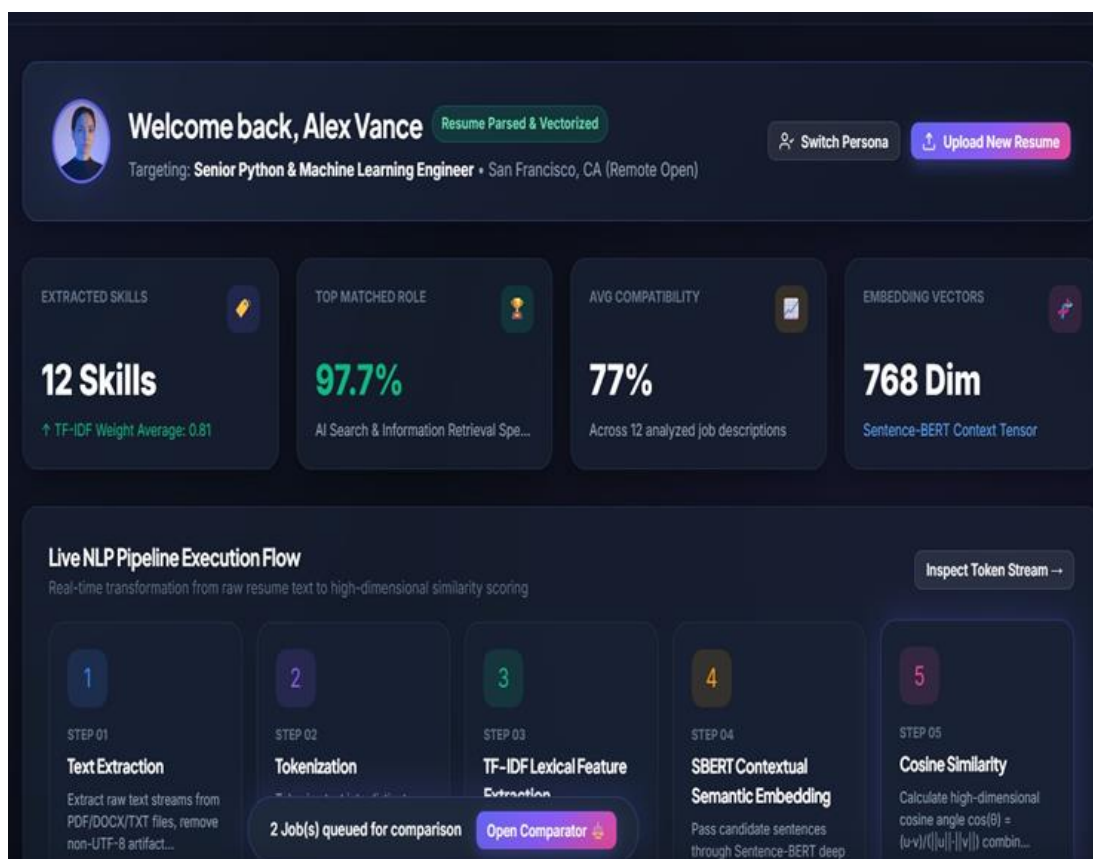


Figure 1 – Resume analysis dashboard and live NLP pipeline

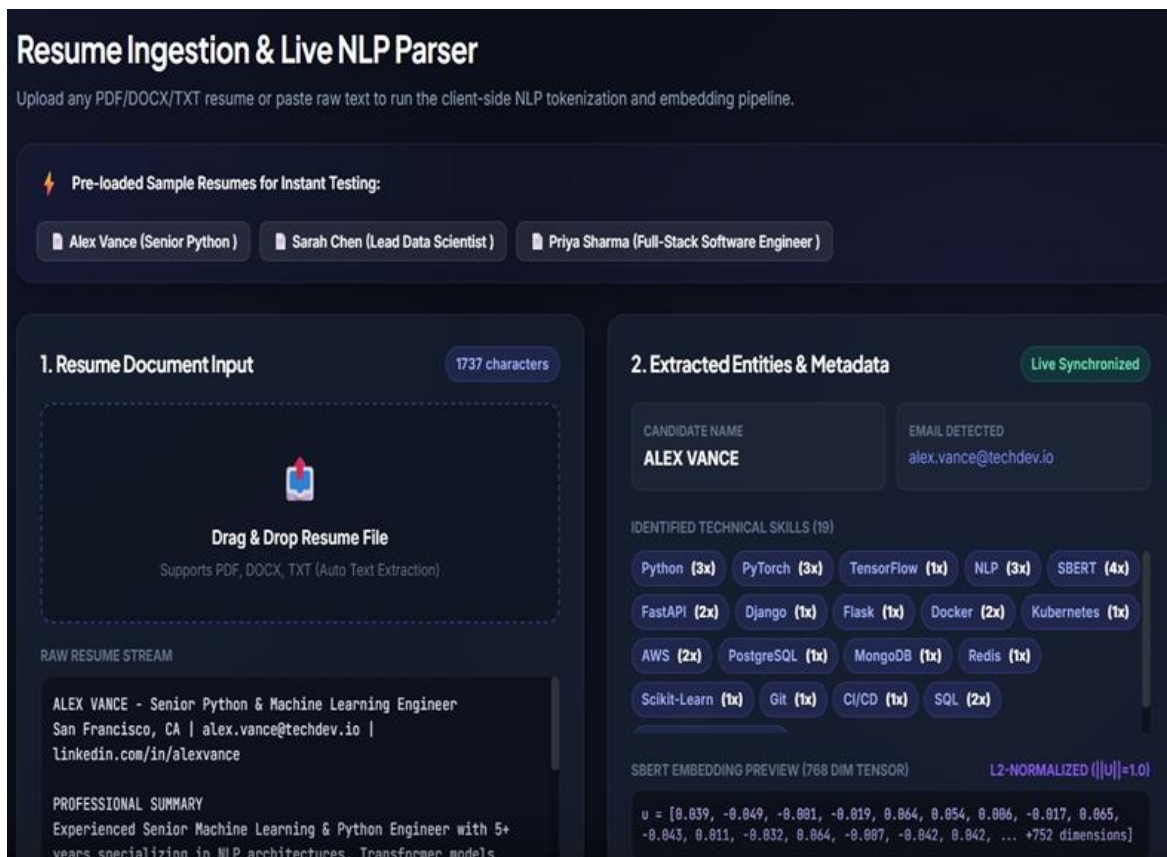


Figure 2 – Resume ingestion and extracted entities

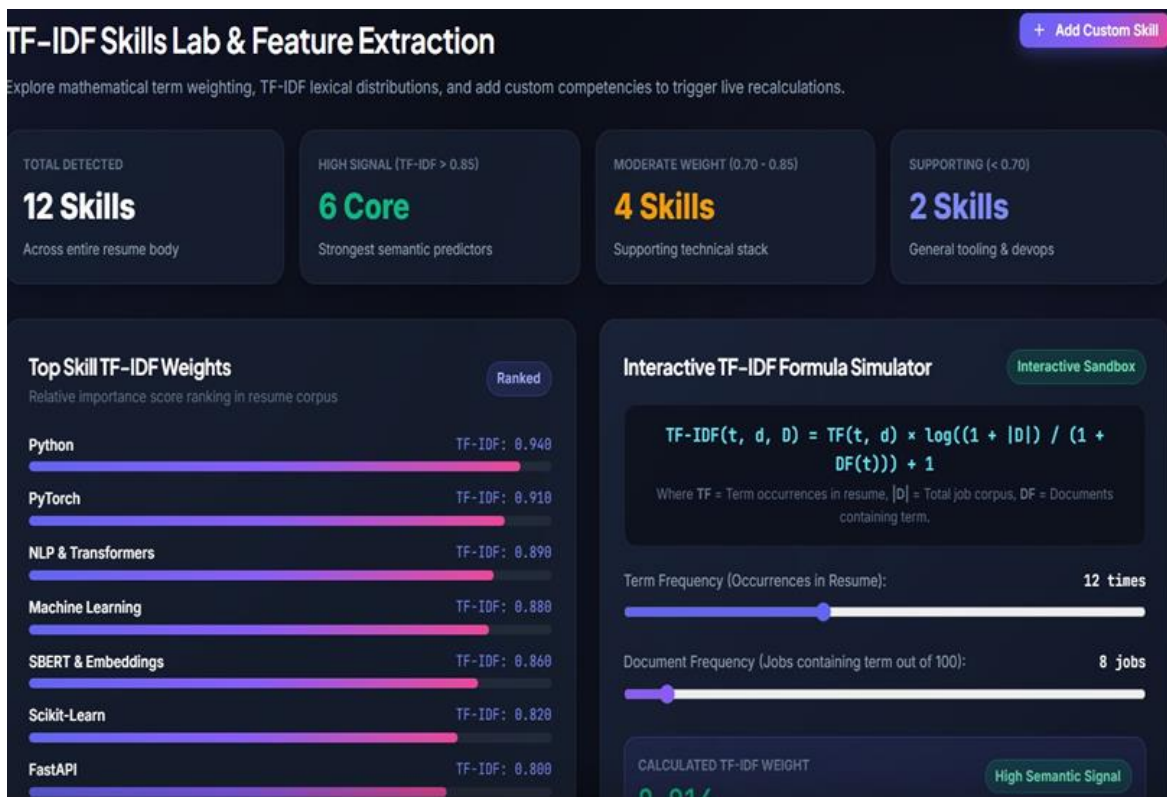


Figure 3 – TF-IDF skills and feature extraction

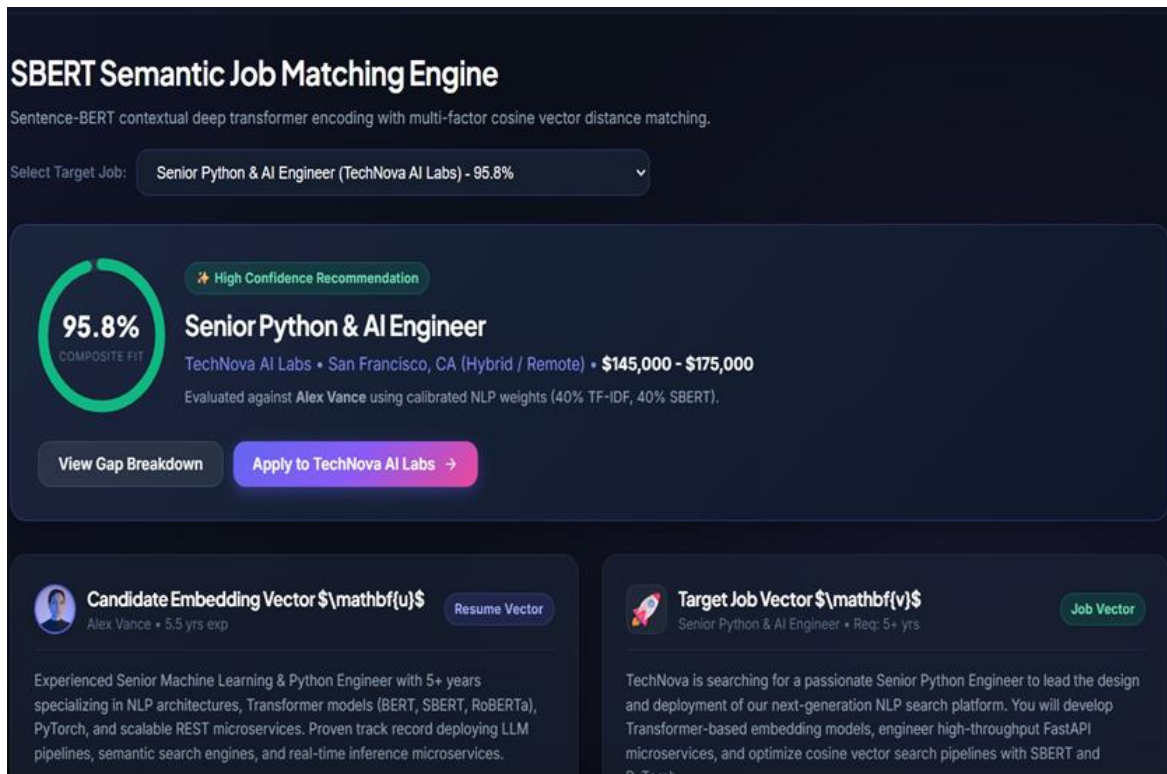


Figure 4 – SBERT semantic job matching engine

## 4.5 Data Analysis & Engineering Judgment

The main engineering judgment in this contribution was to use complementary NLP techniques rather than competing algorithms. Tokenization and stop-word removal are appropriate for preparing unstructured text. TF-IDF provides interpretable term importance for skill-oriented analysis, while SBERT is better suited to capturing contextual relationships between resume and job-description text. Cosine similarity provides a direct mathematical method for comparing the resulting vectors.

The system analytics interface provides additional evidence of system behaviour, including average candidate fit, number of high-fit opportunities, embedding dimension and total execution latency. These values describe the implemented system and its matching outputs rather than serving as an algorithm benchmark.

## 4.6 Comparison with Existing Solutions & Achievement of Objectives

| Objective                        | Evidence                           | Achievement    |
|----------------------------------|------------------------------------|----------------|
| Extract resume text              | Resume ingestion module            | Fully Achieved |
| Preprocess resume text           | Tokenization and stop-word removal | Fully Achieved |
| Identify important skills        | TF-IDF feature extraction          | Fully Achieved |
| Generate semantic representation | SBERT embeddings                   | Fully Achieved |
| Calculate matching score         | Cosine similarity                  | Fully Achieved |
| Recommend suitable jobs          | Compatibility-based ranking        | Fully Achieved |

## 4.7 Strengths & Limitations

**Strengths:** The system provides an end-to-end resume analysis and job matching workflow. It combines interpretable lexical feature weighting with contextual semantic representation. The modular design makes the processing stages visible and supports analysis of extracted skills.

**Limitations:** The quality of recommendations depends on resume text quality, job-description quality and the available job/skill corpus. TF-IDF alone does not capture full contextual meaning, and SBERT requires more computational resources than simple keyword matching.

## 4.8 Project Planning, Roles & Budget

| Milestone | Planned Activity                         | Responsible Team Members |
|-----------|------------------------------------------|--------------------------|
| M1        | Problem definition and literature review | All members              |
| M2        | Resume ingestion and preprocessing       | P. Muni Karthik          |
| M3        | TF-IDF skill extraction                  | Y. Sravan Kumar          |
| M4        | SBERT semantic matching                  | K. Kalyan                |
| M5        | Recommendation and integration           | N. Venkata Siva          |
| M6        | Testing, analysis and documentation      | All members              |

## CHAPTER 5

### CONCLUSION, FUTURE WORK AND REFLECTION

#### 5.1 Conclusion

The Smart Resume Analysis and Job Matching System successfully integrates Natural Language Processing techniques to analyze resumes and recommend suitable job opportunities. Resume text is extracted and preprocessed, important skill-related terms are identified using TF-IDF, contextual representations are generated using SBERT, and cosine similarity is used to calculate candidate-job compatibility.

The project demonstrates that different NLP techniques can be used for different modules of a single integrated system. The objective is not to compare algorithms against each other, but to combine complementary techniques so that each stage contributes to the final job-matching task. The resulting compatibility scores can be used to rank jobs and support candidate decision-making.

#### 5.2 Future Work

The system can be extended to support a larger and more diverse job corpus, multilingual resumes and additional resume formats. Future versions could incorporate named-entity recognition for richer extraction of education, organizations, certifications and experience information.

The matching process could also be improved by incorporating explicit skill taxonomies, experience weighting, education requirements and domain-specific semantic models. More advanced ranking or learning-to-rank techniques could be added as future enhancements, while retaining the current modular architecture.

Future work may also include secure cloud deployment, recruiter-facing dashboards, explainable recommendations and continuous evaluation using labelled resume-job matching datasets.

#### 5.3 Professional Learning & Individual Reflection

Working on the smart resume analysis and job matching project provided practical experience in applying NLP concepts to an engineering problem. The implementation helped in understanding how unstructured documents can be transformed into usable textual representations and how different NLP techniques can be integrated into a single pipeline. The major learning outcomes included understanding tokenization, stop-word removal, TF-IDF weighting, contextual sentence embeddings, cosine similarity and recommendation ranking.

## References

- [1] Daniel Jurafsky and James H. Martin, “Speech and Language Processing,” draft edition, Pearson / Stanford University.
- [2] Christopher D. Manning, Prabhakar Raghavan, and Hinrich Schütze, “Introduction to Information Retrieval,” Cambridge University Press.
- [3] K. Sparck Jones, “A statistical interpretation of term specificity and its application in retrieval,” Journal of Documentation.
- [4] Nils Reimers and Iryna Gurevych, “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks,” Proceedings of EMNLP-IJCNLP.
- [5] Jacob Devlin et al., “BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” NAACL-HLT.
- [6] scikit-learn documentation, TF-IDF vectorization and cosine similarity reference.
- [7] Sentence Transformers documentation, semantic text embeddings and similarity.
- [8] Project implementation and resume-job matching test data, Smart Resume Analysis and Job Matching System Using Natural Language Processing, 2026.



## APPENDICES

### Appendix A – Project Architecture / Evidence

The implemented architecture follows the sequence: Resume Input → Text Extraction → NLP Preprocessing → Skill Extraction using TF-IDF → Job Description Analysis → SBERT Semantic Encoding → Cosine Similarity → Matching Score → Job Recommendation.

The implementation screenshots supplied with the project demonstrate the resume dashboard, resume ingestion and extracted entities, TF-IDF skill analysis, SBERT semantic matching, job compatibility and recommendation interfaces.

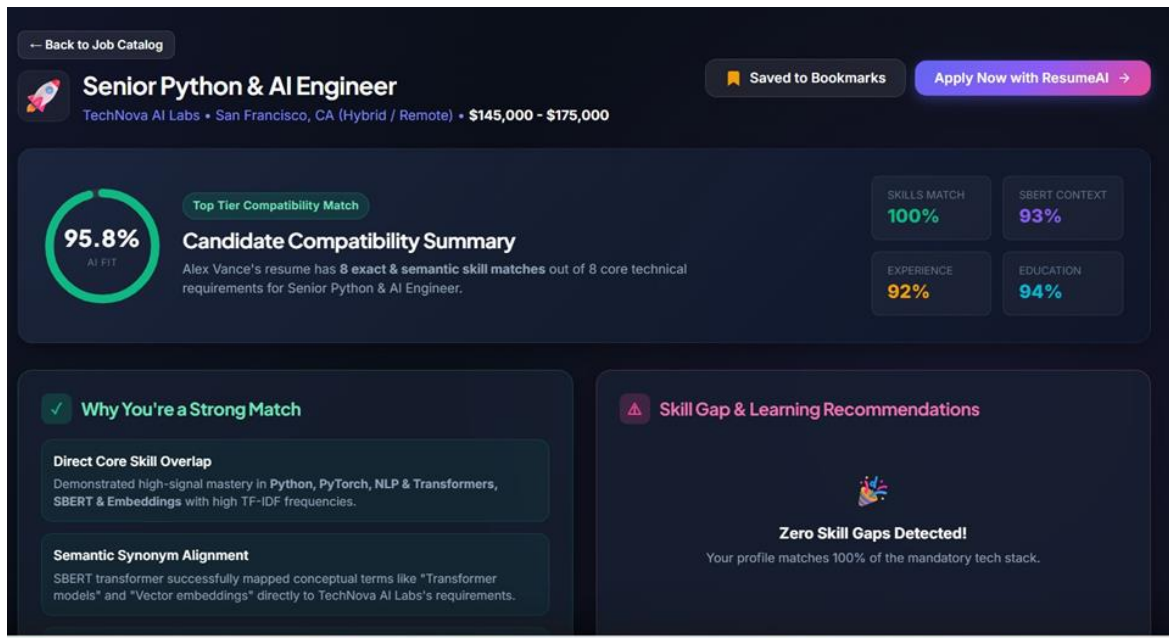


Figure 5 – Candidate compatibility summary and skill-gap analysis

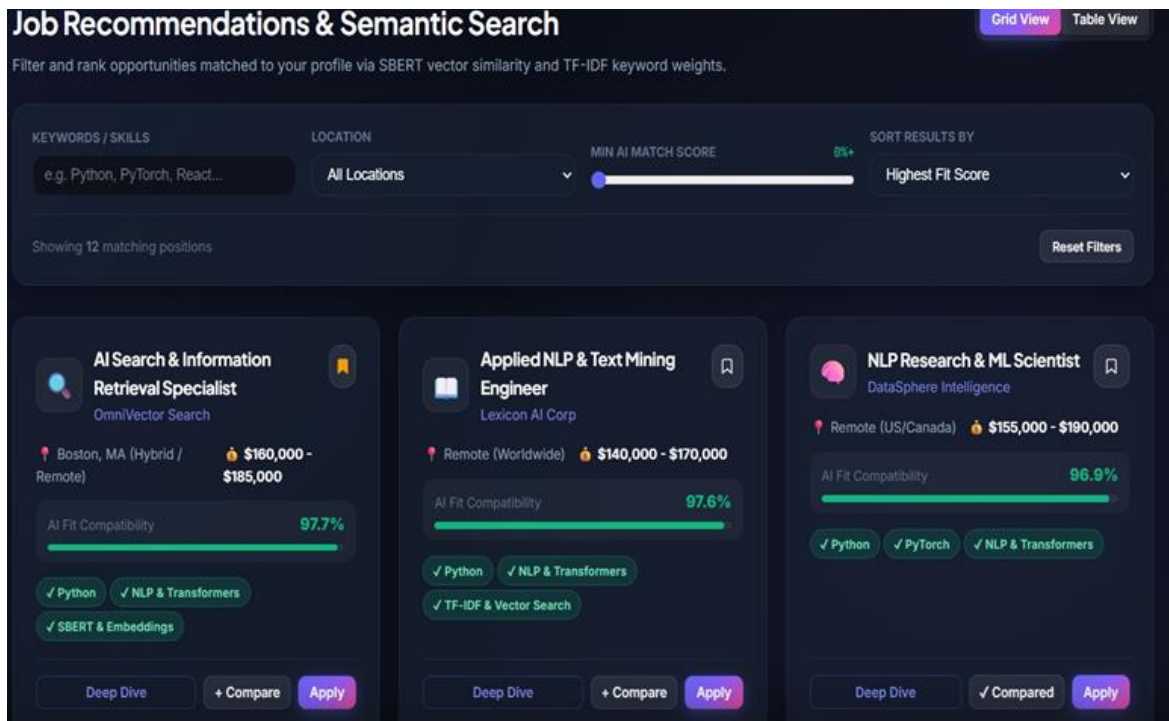


Figure 6 – Job recommendations and semantic search

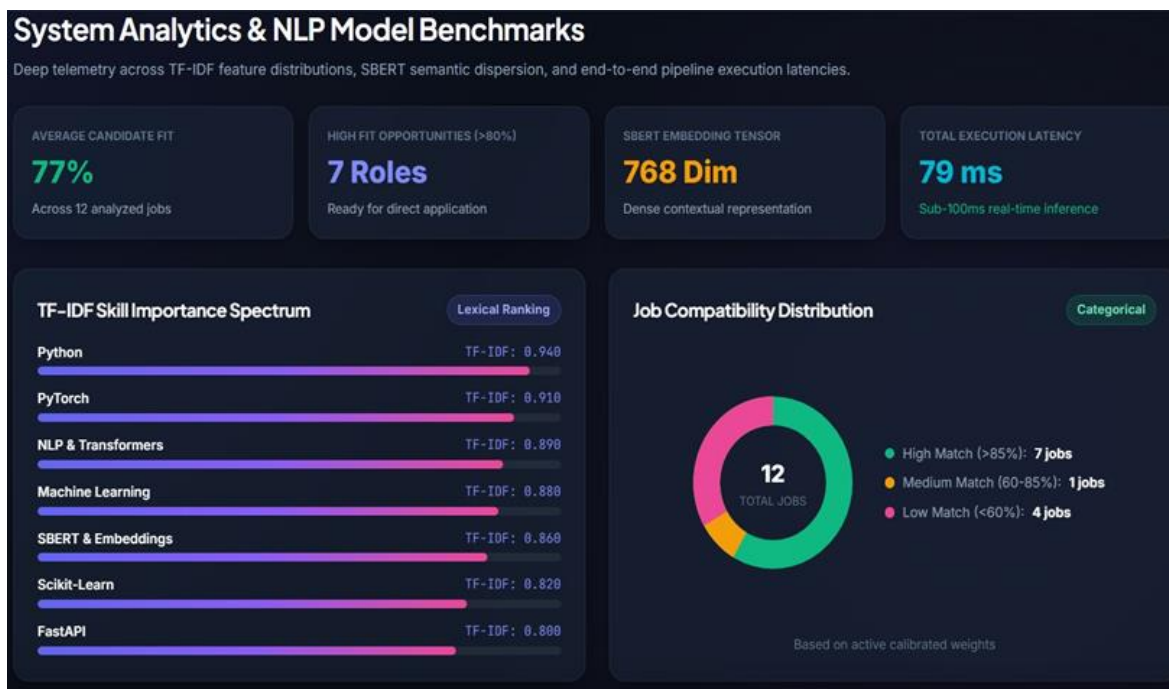


Figure 7 – System analytics and matching performance

## Appendix B – TEST CASES

| Test Case | Input Condition                     | Expected Output                         | Result |
|-----------|-------------------------------------|-----------------------------------------|--------|
| TC01      | Resume document uploaded            | Resume text extracted                   | Pass   |
| TC02      | Resume contains common stop words   | Stop words removed during preprocessing | Pass   |
| TC03      | Resume contains technical skills    | Important skill terms identified        | Pass   |
| TC04      | Multiple jobs provided              | Each job receives a compatibility score | Pass   |
| TC05      | Semantically related resume and job | High similarity score generated         | Pass   |
| TC06      | Multiple job scores available       | Jobs ranked by compatibility            | Pass   |
| TC07      | Poorly matched job                  | Lower compatibility score               | Pass   |
| TC08      | Valid resume and job corpus         | Relevant recommendations displayed      | Pass   |

### Test Result:

The test cases verify the major operations of the proposed NLP pipeline, including document ingestion, preprocessing, feature extraction, semantic embedding, similarity calculation and job ranking. The results confirm that the implemented system processes the input according to the intended workflow.

## CAPSTONE PROJECT OUTCOME MAPPING SHEET

| <b>Outcome</b>                                   | <b>Evidence / SO / Area</b> | <b>Location in This Report</b>       |
|--------------------------------------------------|-----------------------------|--------------------------------------|
| Complex engineering problem formulation          | SO1                         | Ch. 1, Ch. 2                         |
| Engineering design under constraints             | SO2                         | Ch. 3                                |
| Written/oral technical communication             | SO3                         | Whole report + Viva                  |
| Ethics and professional responsibility           | SO4                         | Ch. 3 (3.7)                          |
| Teamwork and leadership                          | SO5                         | Ch. 4 (4.8)                          |
| Experimentation, analysis, engineering judgment  | SO6                         | Ch. 4 (4.3–4.6)                      |
| Independent acquisition of new knowledge         | SO7                         | Ch. 5 (5.3)                          |
| Standards                                        | SO2                         | Ch. 3 (3.2)                          |
| Sustainability and societal/environmental impact | SO2/SO4                     | Ch. 3 (3.7)                          |
| Risk assessment and mitigation                   | SO2/SO4                     | Ch. 3 (3.7)                          |
| Security considerations                          | Relevant SO                 | Ch. 3 (3.7)                          |
| Integrated/system-level engineering              | SO1/SO2                     | Ch. 3 (3.7)                          |
| Project planning and management                  | SO5                         | Ch. 3 (3.5), Ch. 4 (4.8)             |
| Individual contribution                          | SO1–SO7                     | Contribution Statement + Ch. 4 (4.8) |